



UNIVERSITY OF RUHUNA
BACHELOR OF INFORMATION AND COMMUNICATION TECHNOLOGY
ICT4163 - Digital Image Processing



Practical 01 – Introduction to Digital Image Processing with Python

- ***Note that: Download the relevant resources for this practical from the resources folder of the LMS.***
- 1. Import the “cv2” (open cv) library and the “cv2_imshow” patch to the google collab.
- 2. Read and display the given image (image_1.jpg).
- 3. Convert the given image (image_1.jpg) to a grayscale image and display.
- 4. Resize the given image (image_1.jpg) according to the following width and height of pixels.
 - Width 600 pixels
 - Height 400 pixels
- 5. Rotate the given image (image_1.jpg) by 180 degrees and display.
- 6. Blurring the given Image using “Gaussian Blur” according to the following kernel size and the standard deviation.
 - Kernel size 21,
 - Standard Deviation : let it to select automatically
- 7. Detect the edges of the given image (image_1.jpg) using Canny edge detector, the lower and upper threshold values are as follows.
 - Lower Threshold: 100
 - Upper Threshold: 200
- 8. Draw a rectangle on the given image (image_1.jpg) according to the following values and write your index number and the name inside that.

The rectangle need to be started from the top-left corner of the given image according to the given coordinates (x=50, y=50).

 - Width: 450px
 - Height: 200px
- 9. Threshold the given image (image_1.jpg) using the following threshold value.
 - Threshold Value: 130
 - The maximum value that a pixel can take when it is above the threshold: 255